

ME 300 Thermodynamics – Summer 2021

Exam 2

18:30

Part I) _____ (30 pts)

(Part II) _____ (40 pts)

(Part III) _____ (30 pts)

TOTAL _____

Sauntions

Directions

1. Open the online answers portion of the exam and download the exam (this document).
2. Make sure to show all your work!
 - a. *If you are using a tablet*, solve the problems directly on this document, showing all your work and writing as clearly as possible.
 - b. *If you are writing on paper*, solve the problems on a separate piece of paper (you do not have to print the exam), showing all your work and writing as clearly as possible.
3. Questions that require answers to be submitted to the online answers page on Canvas are written in *red italics* below as a way to remind you of what should be entered. All sketches and plots should be clearly drawn on your working paper and submitted with the upload.
4. If you have questions, please use the private chat function in Zoom to send a message directly to Dr. O'Connor so as not to disturb the other students. DO NOT UNMUTE AND ASK YOUR QUESTION ALOUD – this will be distracting to the other students.
5. Once the answers portion of the exam closes, scan or take very nice pictures of your work and upload it to the upload portion of the exam in a PDF document. I trust that you will not change anything on your exam after the answers portion of the exam closes. I can't enforce this – I'm really just relying on you to do the right thing. Please keep to the honor code.
6. The timeline for the exam will be the following (all times in EDT, Friday, July 23):
 - a. Exam opens (answers portion and upload portion): 11:00 AM
 - b. Answers portion of the exam closes: 12:30 PM
 - c. Upload portion of the exam closes: 12:45 PM
7. You can anonymously report any incidents of cheating using the following Google form:
<https://forms.gle/E4wFjDtvNuginVkp9>

Academic Integrity Pledge

I pledge that I am adhering to Penn State's Code of Conduct when taking this exam.

- I will not communicate with anyone during the exam except Dr. O'Connor.
- I will use only the allowed materials (calculator, notes, tables, and book), I will only access the internet to download the exam, view the e-book and notes on Canvas (no videos), and to submit my answers and upload the exam.
- I will not change any of my written answers after the exam but before I upload my work.
- I will not cheat in any way.

It is important to me to be a person of integrity and that means that all answers on this exam are my individual answers. *In the online answers, please type your full legal name to agree to the academic integrity pledge.*

Part I: Concept Questions

1. (10 points) Can the thermal efficiency of a heat engine ever reach one? *Answer the multiple-choice question in Canvas.*

Based off Spl 7 Practice Exam, #I.1

YES

NO ⁺²

Explain your response to this question. *Type your answer in the box in Canvas.*

2nd law says (Kelvin-Planck) that heat cannot be completely converted to work. So if $\eta_{th} = \frac{\bar{W}_{net}}{\bar{Q}_{in}}$ and $\bar{W}_{net} < \bar{Q}_{in}$, then $\eta_{th} < 1$ ⁺³

2. (10 points) Identify the correct sign of the change in entropy for the following three processes.

- a) Fluid undergoes a reversible cooling process. Is the change in entropy: *Answer the multiple-choice question in Canvas.*

Based off quiz #14 and book table #7.2

GREATER THAN ZERO

LESS THAN ZERO ⁺²

EQUAL TO ZERO

- b) An ideal gas undergoes an irreversible heating process. Is the change in entropy: *Answer the multiple-choice question in Canvas.*

GREATER THAN ZERO ⁺²

LESS THAN ZERO

EQUAL TO ZERO

- c) An ideal gas undergoes an adiabatic and reversible expansion. Is the change in entropy: *Answer the multiple-choice question in Canvas.*

GREATER THAN ZERO

LESS THAN ZERO

EQUAL TO ZERO ⁺²

- d) Use an equation to explain the answers to these previous three questions. *Type your answer in the box in Canvas.*

$\Delta S' = \oint \frac{\delta Q}{T} + \Phi_2$ by Clausius's entropy statement, where Φ_2 is entropy generation due to irreversibilities and must be > 0 ⁺³

3. (10 points) An engineer develops a new surface coating for the blades of a water pump that reduces friction and turbulent mixing in the pump. For a given pressure ratio of the pump, answer the following questions:

- a) Does the isentropic efficiency of the pump increase or decrease with the new coating? *Answer the multiple-choice question in Canvas.*

Based of lecture M7.14

INCREASE

+2

DECREASE

- b) Does the work required to drive the pump increase or decrease with the new coating? *Answer the multiple-choice question in Canvas.*

INCREASE

DECREASE

+2

- c) Explain your answer to the previous two questions. Use an equation to support your discussion. *Type your answer in the box in Canvas.*

$$\eta_{\text{comp}} = \frac{\dot{W}_{\text{ideal}}}{\dot{W}_{\text{real}}} \quad +3 \quad \text{because it takes less work to compress}$$

a fluid in an ideal pump because there are no

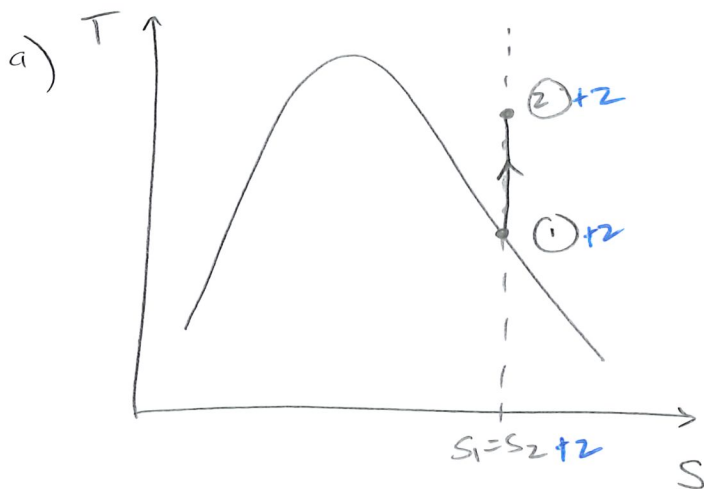
irreversibilities to overcome. ⁺³ If friction/mixing, which are

irreversibilities, are reduced, then $\eta_{\text{comp}} \uparrow$ and $\dot{W}_{\text{real}} \downarrow$

Part II: Short Problems

1. (20 points) Steam is compressed isentropically from a saturated vapor at 2.1 MPa to a pressure of 6 MPa. Identify any tables you use at each step of the problem. If you use a superheated vapor or subcooled liquid table, do not interpolate between values – use the closest value on the table.

- a) (4 points) Draw this process on a T-s diagram. *Draw the figure below or on a separate page – do not enter anything into the Canvas quiz.*
- b) (6 points) What is the temperature at the end of this compression?. *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- c) (10 points) What would the final temperature be if it was air being compressed from the same starting temperature and pressure to the same final pressure? Assume the following properties for air: $c_p = 1000 \text{ J/kg-K}$ and $c_v = 714.29 \text{ J/kg-K}$. *Show your work below or on a separate page and enter this value in the Canvas quiz.*



b) State 1: $P_1 = 2.1 \text{ MPa}$
 $x_1 = 1$ } Table D.2 $\rightarrow S_1 = 6.3210 \frac{\text{kJ}}{\text{kg-K}}$
 $T_1 = T_{\text{sat}} = 488.01 \text{ K}$

State 2: $P_2 = 6 \text{ MPa}$
 $S_2 = S_1 = 6.3210 \frac{\text{kJ}}{\text{kg-K}}$ } Table D.3N $\rightarrow S_2 = 6.3211 \frac{\text{kJ}}{\text{kg-K}}$
 $\Rightarrow T_2 = 620 \text{ K}$

c) For isentropic:

$$T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} = 488.01 \left(\frac{6}{2.1} \right)^{\frac{0.4}{1.4}} \Rightarrow T_2 = 658.7 \text{ K}$$

blank

2. (20 points) A reversible Carnot cycle operates between 1200 K and 300 K with the following four steps:

- Process 1-2: isentropic compression
- Process 2-3: isothermal heat addition
- Process 3-4: isentropic expansion
- Process 4-1: isothermal heat rejection

Based on example in
video M7.8

- a) (6 points) What is the thermal efficiency of this cycle? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- b) (6 points) If 10 kJ of heat is rejected from the cycle in Process 4-1, how much heat was added in Process 2-3 in [kJ]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- c) (8 points) What is the change in entropy of the working fluid for each process in the cycle in [J/K]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*

a) reversible cycle $\rightarrow \eta_{th} = \eta_{Carnot} = 1 - \frac{T_L}{T_H}$

$$= 1 - \frac{300}{1200} \rightarrow \boxed{\eta_{th} = 0.75}$$

b) $Q_{out} = 10 \text{ kJ}$

1st law: $Q_{in} = \bar{W}_{net} + Q_{out}$ and $\eta_{th} = \frac{\bar{W}_{net}}{Q_{in}}$

$\bar{W}_{net} = \eta_{th} Q_{in}$

$Q_{in} = \eta_{th} Q_{in} + Q_{out}$

$Q_{in} (1 - \eta_{th}) = Q_{out}$

$Q_{in} = \frac{Q_{out}}{1 - \eta_{th}} = \frac{10 \text{ kJ}}{1 - 0.75} \rightarrow \boxed{Q_{in} = 40 \text{ kJ}}$

c) $\Delta S' = \int \frac{\delta Q}{T}_{rev}$ for each process

$\rightarrow$ Process 1-2: $\Delta S' = 0$ b/c isentropic

$$\rightarrow \text{Process 2-3: } \Delta S' = \int \frac{\delta Q}{T} \Big|_{\text{rev}} \quad +2$$

$$= \frac{Q_{\text{in}}}{T_H} = \frac{40 \text{ kJ}}{1200 \text{ K}} \rightarrow \boxed{\Delta S' = 33.3 \frac{\text{J}}{\text{K}}}$$

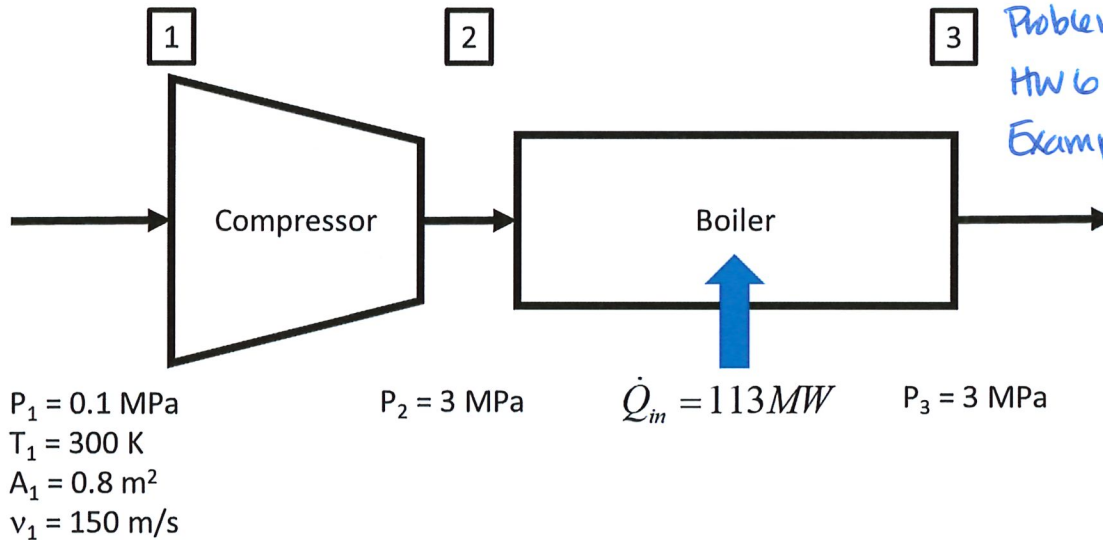
$$\rightarrow \text{Process 3-4: } \Delta S' = 0 \quad \text{b/c isentropic} \quad +1$$

$$\rightarrow \text{Process 4-1: } \Delta S' = \int \frac{\delta Q}{T} \Big|_{\text{rev}} \quad +2$$

$$= \frac{Q_{\text{out}}}{T_L} = \frac{-10 \text{ kJ}}{300 \text{ K}} = \boxed{-33.3 \frac{\text{J}}{\text{K}}}$$

Part III: Analysis (30 points)

The story: A factory needs steady high-pressure, heated air for a fabrication process. In order to provide this air, the factory has an ideal air compressor followed by a boiler, as shown in the drawing below. Do the following steps of the analysis to calculate different properties for these processes. Use the following constant properties of air: $R=287 \text{ J/kg-K}$, $c_p=1001 \text{ J/kg-K}$, $\gamma=1.4$.



- (5 points) What assumptions must be made about both the working fluid and the devices in this factory in order to solve for the state points? *Enter these in the Canvas quiz – you do not need to also write them on the page.*
- (6 points) What is the mass flow through the system in [kg/s]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- (6 points) What is the temperature of the air after the compressor in [K]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- Waste to drive compressor
- (6 points) What is the temperature of the air after the boiler in [K]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- (7 points) What is the change in entropy of the air through the boiler in [J/kg-K]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*

a) Air $\rightarrow$ ideal gas $^{+1/2}$
 $\rightarrow$ const. c_p, c_v $^{+1/2}$

Compressor $\rightarrow$ steady-flow $^{+1/2}$
 $\rightarrow \Delta ke = \Delta pe = 0$ $^{+1/2}$

$\rightarrow \dot{Q} = 0$ $^{+1/2}$
 $\rightarrow$ reversible (ideal) $+1$

Boiler $\rightarrow$ steady-flow $^{+1/2}$

$\rightarrow \Delta ke = \Delta pe = 0$ $^{+1/2}$

$\rightarrow \dot{W} = 0$ $^{+1/2}$

$$b) \dot{m} = \rho v A = \rho_1 v_1 A_1 \quad +2$$

$$\rho_1 = \frac{P_1}{RT_1} = \frac{100000}{(287)(300)} = 1.161 \frac{\text{kg}}{\text{m}^3} \quad +1$$

$$\dot{m} = \rho_1 v_1 A_1 = (1.161)(150)(0.8) \rightarrow \boxed{\dot{m} = 139.2 \frac{\text{kg}}{\text{s}}} \quad +1$$

$$c) \dot{Q} = 0 + \text{reversible} \rightarrow \text{isentropic} \quad +2$$

$$T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} = 300 \left(\frac{3}{0.1} \right)^{\frac{0.4}{1.4}} \rightarrow \boxed{T_2 = 792.79 \text{ K}} \quad +1$$

$$d) \dot{m}(\Delta h + \Delta \cancel{v_e} + \Delta \cancel{v_e}) = \dot{Q} - \dot{W} \quad +2$$

$$\dot{m} \Delta h = \dot{Q} \quad +2$$

$$\dot{m} c_p \Delta T = \dot{Q} \rightarrow T_3 = \frac{\dot{Q}}{\dot{m} c_p} + T_2$$

$$= \frac{(113 \times 10^6)}{(139.2)(1001)} + 792.79$$

$$\boxed{T_3 = 1603.8 \text{ K}} \quad +1$$

$$f) \text{ isobaric process} \quad +2$$

$$\Delta S = c_p \ln \left(\frac{T_2}{T_1} \right) - R \ln \left(\frac{P_2}{P_1} \right) \quad +3$$

$$= (1001) \ln \left(\frac{1603.8}{792.79} \right) \rightarrow \boxed{\Delta S = 705.28 \frac{\text{J}}{\text{K}}} \quad +1$$

$$d) \dot{m}(\Delta h + \Delta \cancel{v_e} + \Delta \cancel{v_e}) = \dot{Q} - \dot{W} \quad +2$$

$$\dot{W} = -\dot{m} \Delta h = -\dot{m} c_p \Delta T = -(139.2)(1001)(792.79 - 300)$$

$$\boxed{\dot{W} = -687.14 \text{ MW}} \quad +1$$

